

Blue Carbon Guardian

Tide-aware satellite monitoring for Gulf mangroves — from alert to evidence.

Team Blue Athar • الأثر الأزرق

Theme: Ecosystem Health, Biodiversity & Blue Carbon

91 stands

Abu Dhabi + Tarut
Bay

5 confirmed

real conversions, 0.3–0.5
m

73 % vs 13 %

partial loss caught vs chance

≈1

false alarm /
stand-year

Nahla Nabil • Bahrain

LIVE MONITOR • ABU DHABI PILOT



■ conversion confirmed on sub-metre imagery

Restoration is scaling faster than monitoring

 **100 M**

mangroves pledged by the UAE by 2030

COP26, 2021 · Abu Dhabi Mangrove Initiative, 2022

 **100 M+**

mangroves pledged by Saudi Arabia by 2030

Saudi Green Initiative

 **Feb 2026**

first GCC mangrove monitoring guide

EAD + IUCN Mangrove Specialist Group

TODAY

Periodic field surveys and one-off studies; a 2025 regional review calls Gulf monitoring **“uneven”**. No standing alert for each stand, no independent check of carbon claims.

THIS IS HAPPENING NOW · STAND 5, ABU DHABI



Sentinel-2 true colour, contains modified Copernicus Sentinel data. Our alert began on 25 Oct 2022, during the works.

SDG 13

SDG 14

SDG 15

Sources: ead.gov.ae · admangroves.ae · spa.gov.sa · Frontiers in Marine Science 2025 (doi:10.3389/fmars.2025.1695426)

Who uses it, and for which decision



Environment agencies

EAD, MOCCA, Saudi NCVC

THEIR DECISION

Which stands do we inspect this month? Is a protected stand intact?

Today: Periodic field surveys, one-off studies

WITH BLUE CARBON GUARDIAN

Monthly alert + 200 m cell map: Inspect only the flagged cells



Planting & carbon-project developers

ADNOC, Red Sea Global, carbon developers

THEIR DECISION

Is the carbon stock we report still there?

Today: Field plots and drone surveys, campaign by campaign

WITH BLUE CARBON GUARDIAN

Condition + carbon stock with a range per stand, a site report for verification



Coastal developers & regulators

EIA consultants, permit offices

THEIR DECISION

Did construction stay outside the mangrove line?

Today: Comparing two dated images by eye

WITH BLUE CARBON GUARDIAN

An alert when works enter mangrove cells, with dated evidence



New scene

every ~5 days



Alert

~1 false alarm / stand-yr



Cell map

where to look



Inspection

only flagged cells



Site report

carbon at stake

91 mangrove stands, six years of free imagery

PILOT: 25 STANDS · 2,281 HA · ABU DHABI



Outlines: WorldCover 2021 patches ≥ 3 ha. Red: the five confirmed conversions. **+66 unseen stands** (Abu Dhabi N/W, Saudi Tarut Bay) test transfer.



Sentinel-2 L2A

ESA / Copernicus · time series, 10–20 m

free · 778 dates



EnMAP

DLR · 224 bands · independent hyperspectral check

free · 2 scenes



ESA WorldCover 2021

10 m land cover · stand outlines

CC BY 4.0



Field carbon plots

Schile et al. 2016 · carbon density

CC0 · 24 plots



Sub-metre captures

WorldView-2/3, Legion-1 · confirming conversions

viewed only (Esri Wayback)

Sentinel-2 · 778 dates

EnMAP · 224 bands

Sub-metre captures

Field carbon plots



From satellite pixels to an inspection decision



CHECKS BUILT IN, INDEPENDENT OF THE MODEL

EnMAP hyperspectral
same dates, $r = 0.84\text{--}0.99$

Simulated losses
every rate vs its chance rate

Sub-metre captures
5 conversions confirmed

65 unseen stands
0.79 false alarms / stand-yr

PROBLEM STATEMENT

“We monitor mangrove condition and carbon stock in Abu Dhabi, 2020–2026, so that authorities and developers can decide where to inspect and verify carbon claims.”

The tide looks like tree loss – until you correct for it

WHY NAIVE MONITORS FAIL ON GULF TIDAL FLATS

same trees, same day of the year

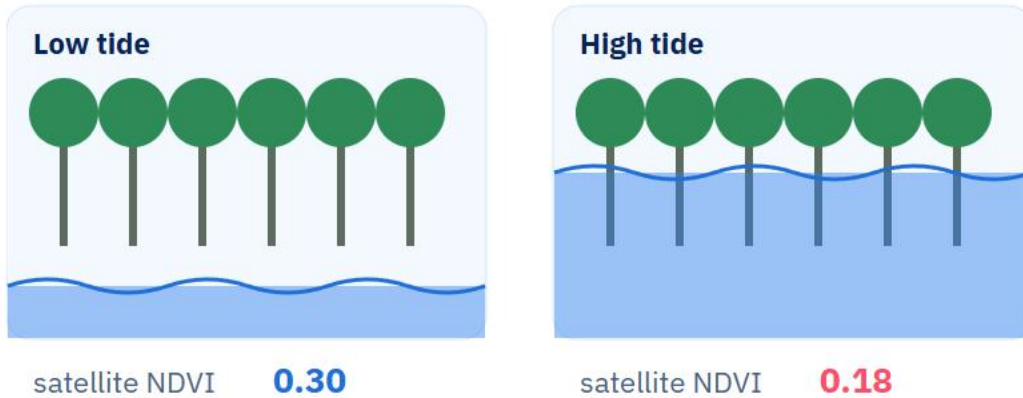
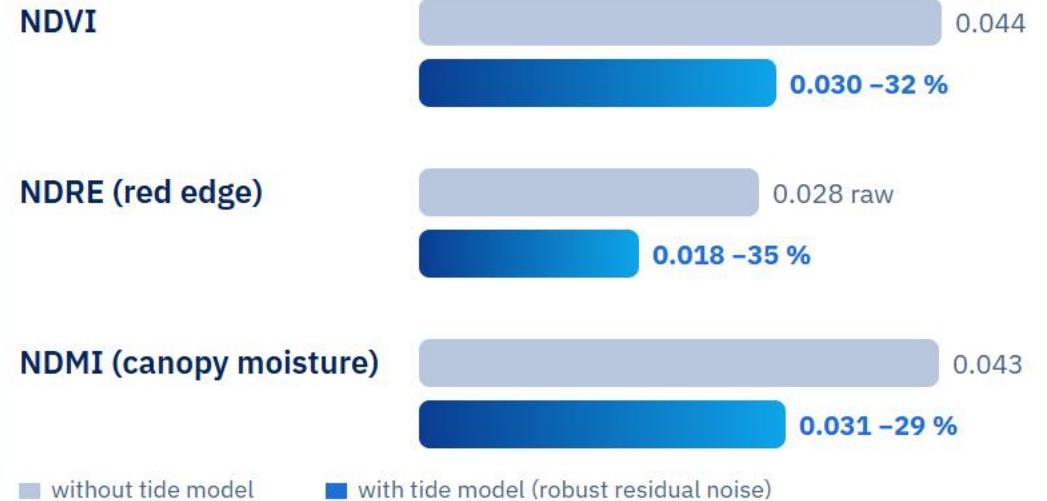


Illustration. Measured on our stands: index noise vs wetness, $r \approx -0.9$.

DATE-TO-DATE NOISE, 25 STANDS, 2020-2026



We regress each index on trend, season **and tidal state** before flagging anything: **29–35 % less noise** on every index.

A stand average hides partial loss. 200 m cells do not.



Stand 5, winter 2022–23 (real data): 200 m cells turn red while the stand average barely moves. Contains modified Copernicus Sentinel data.

8 of 28

cells of stand 9 in severe decline while its average reads “stable”

3 of 5

real conversions found **only** by the cells (stands 12, 18, 6)

ONE ALERT RULE

stand $\leq -1.5\sigma$ OR any cell $\leq -3.5\sigma$

calibrated together: 0.96 false-alarm episodes per stand-year

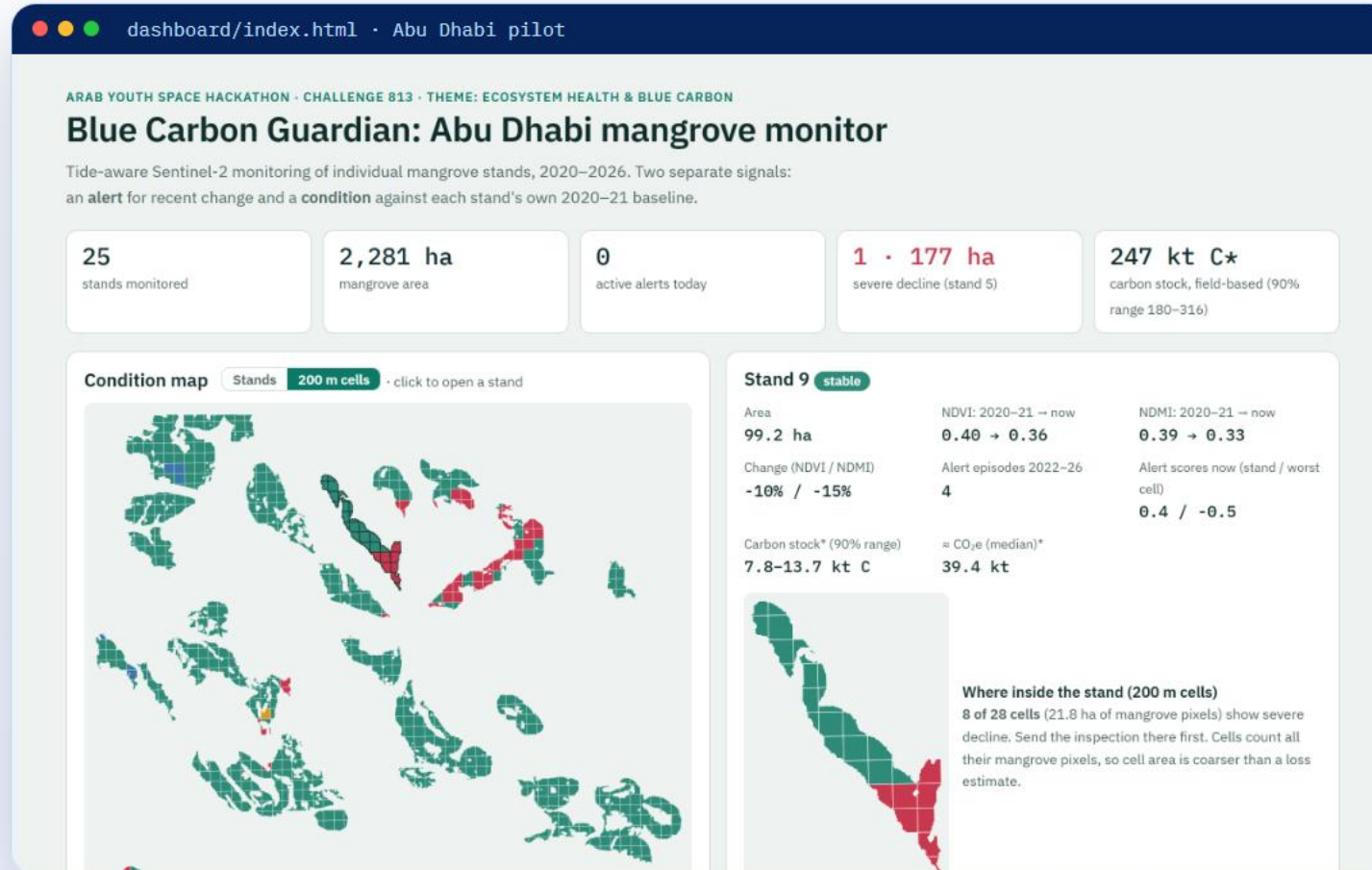
**Partial loss caught within 60 days
(simulated)**

73 % stand + cells

25 % stand average alone

13 % by chance

From signal to decision, in one screen



- One alert**
stand average OR any 200 m cell, ~1 false alarm per stand-year
- Where to look**
a 200 m cell map for the inspector
- Condition**
last 12 months vs the stand's own 2020–21 baseline
- Carbon at stake**
stock with a range, from field data
- Site report**
one printable page per stand; every sentence traces back to a number

dashboard

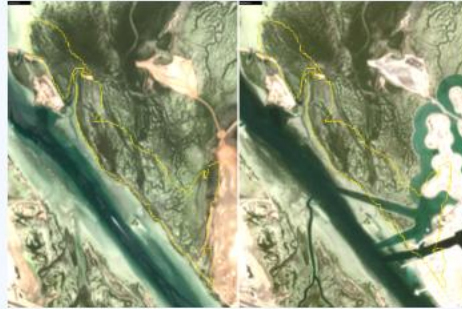
site reports

JSON alert API (concept)

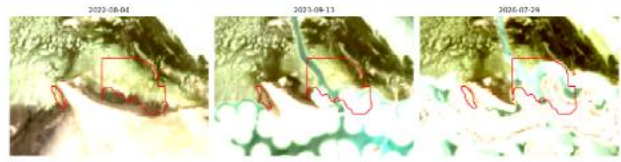
Five real conversions, confirmed on sub-metre imagery



Stand 5 · reclamation with finger canals



Stand 9 · canal development takes the south-east



Stand 12 · found only by the cells



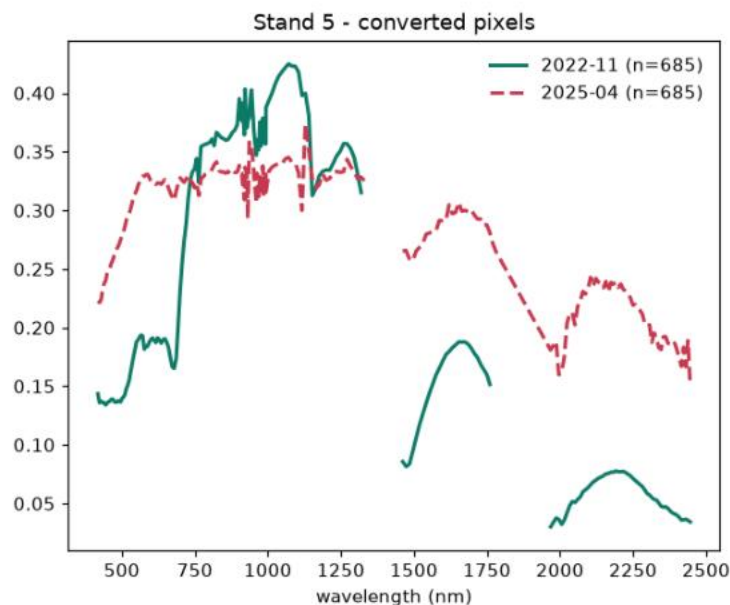
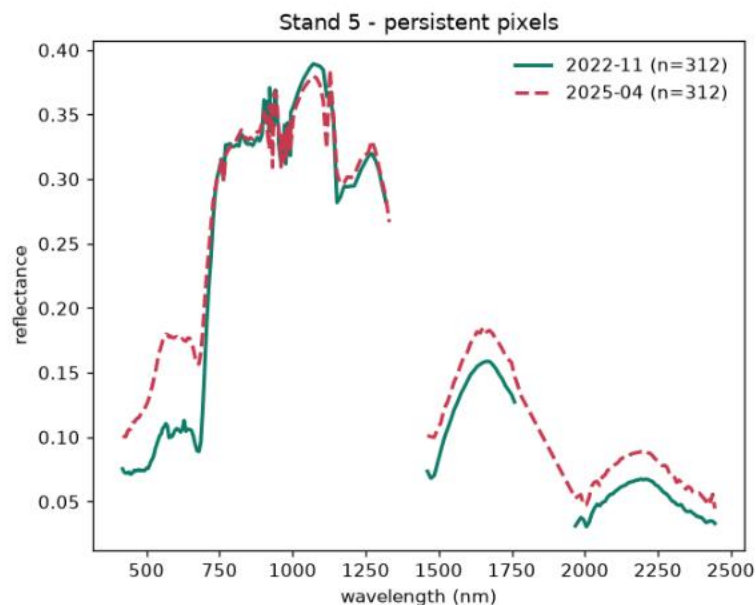
Stand 18 · found only by the cells, 2025

WHEN DID THE ALERT START?



Dated 0.3–0.5 m captures (WorldView-2/3, Legion-1) confirm all five, including stand 6 where a lagoon replaced part of the stand. At every site the alert began **between the last intact and the first disturbed capture**: detection of works as they happen, not advance warning.

An independent check, and a material fingerprint



Stand 5 on EnMAP 2022 grid: converted (red) / persistent (green)



Contains modified EnMAP data (c)DLR 2022, 2025



$r = 0.84-0.99$

EnMAP vs our Sentinel-2 indices, same dates,
25 and 14 stands



62 vs 26 ha

stand 5 converted: EnMAP pixels vs our
conservative Sentinel-2 flag



F1 0.74 vs 0.72

full spectrum vs Sentinel-2-like bands as a
classifier: no real gain, and we say so

Converted pixels change from a vegetation spectrum to a bright mineral-like one (sand and fill): reclamation, not dieback. Contains modified EnMAP data © DLR 2022, 2025.

Can we detect a loss? Tested against pure chance

ARTIFICIAL LOSS INJECTED INTO REAL STANDS WITH NO KNOWN CHANGE

A detection counts only if a **new** alert starts after the loss begins. The same test with **no** loss gives the chance rate. ~160–240 trials per row (±7 points).



CALIBRATED • 25 PILOT STANDS

0.96

false-alarm episodes per stand-year (117 stand-years)

UNSEEN • 65 STANDS, THRESHOLDS FIXED

0.79

per stand-year over 305 stand-years, incl. Saudi Tarut Bay



We corrected our own first test: it counted alerts that were already running and had no chance baseline (it reported 58 % / 90 %). We withdrew it.

Where it breaks — and what we do not claim



Detection, not early warning

Alerts fire as works happen. We never claim to see stress before it is visible.



10 m pixels

Patches under ~0.6 ha and young plantings are missed, Bahrain's small stands included.



Imagery, not site visits

Conversions confirmed on dated sub-metre captures, not yet on permits or field records.



Simulated losses

Canopy-cover loss, not leaf stress; every rate next to its chance rate.



Hyperspectral

EnMAP confirms our indices but was not a better classifier (F1 0.74 vs 0.72).



Carbon

Area × field density from four sites (2013–14): a stock with a range, never a sequestration rate.

Rule we kept: no detection rate without its chance rate; no claim of advance warning, field verification or species discrimination.

Carbon stock, grounded in field measurements

108 t C/ha

local mean density (90 % CI 79–138), 24 plots at 4 sites

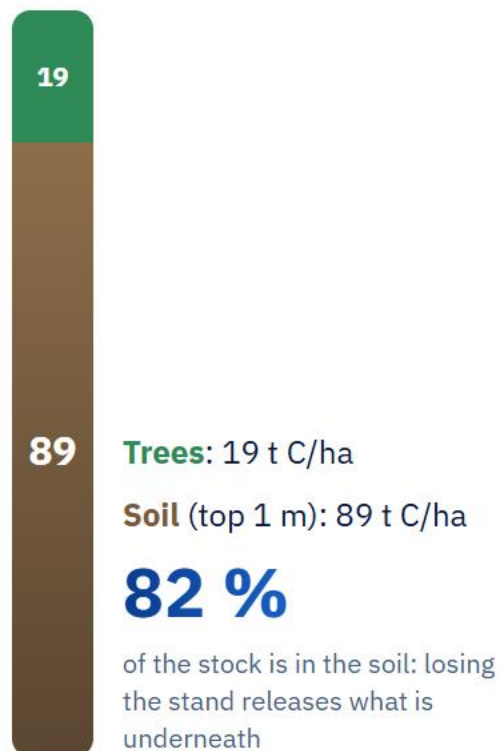
247 kt C

pilot stock, 2,281 ha (90 % range 180–316)

≈ 907 kt CO₂e

pilot equivalent, median

WHERE THE CARBON IS



STAND 5 CONVERSION

2.8–6.7 kt C

stock on the 26–62 ha converted (two methods); about USD 0.28–0.66 M at USD 27/t CO₂e, illustration only

Tested, and too weak

Sentinel-2 indices vs field tree carbon (95 plots): best $p = 0.38$. So carbon is **area × field density**, not imagery biomass.

A monitoring service with honest unit economics

MONITOR

\$12k

per site / year ($\approx$ \$5 / ha)

alert, cell map, condition, carbon, monthly reports

EVIDENCE

\$25k

per site / year

+ sub-metre check of every alert, audit pack

DEVELOPMENT WATCH

\$8k

per project / year

for coastal developers and EIA consultants

COST TO SERVE $\approx$ \$3.3K / SITE / YR

Compute < \$5 (measured) · **VHR on alert**
\$500–750 · **Analyst** \$2k (assumption)

REVENUE SCENARIO (HYPOTHESIS)



Customers

Agencies (EAD, MOCCA, NCV) · planting programmes: ADNOC (10 M), Red Sea Global (50 M) · carbon developers · coastal developers and EIA consultants

Competition

Global Mangrove Watch: free monthly loss alerts (Gulf coverage not confirmed). We are complementary: tide-aware, calibrated, stand + cell, carbon and reports.

THE HONEST MARKET

Gulf + Red Sea $\approx$ 16 sites $\rightarrow$ \$0.2–0.4 M / yr: a niche. The real potential: a **national MRV contract**, a monitoring layer on **gIQ**, licensing.

Prices and analyst time are hypotheses for a pilot to test. All 13 sources: docs/business_plan_draft.md.

From proof of concept to a national monitoring layer



Now · PoC

OCT 2026

- Stand + 200 m cell monitor
- dashboard & site reports
- field-grounded carbon
- 5 conversions confirmed
- EnMAP check
- 0.79 false alarms on 65 unseen stands



Incubation

OCT 2026 - JAN 2027

- Satellite 813 in the spectral step
- deployment on gIQ
- a pilot customer
- match alerts to permits & site records
- more EnMAP dates per season



Scale

2027 +

- Saudi Arabia (Gulf & Red Sea) and Bahrain
- very-high-resolution data for small stands
- national MRV contract
- licensing to platforms

Two limits we are designing around, on purpose

Bahrain's own stands are too small for 10–30 m pixels (a case for VHR at incubation) · our five events are confirmed on dated imagery, not yet on site records (the first pilot task).

TEAM BLUE ATHAR · الأثر الأزرق

Every lost mangrove leaves a trace. We find it from space.

BUILT END TO END BY ONE ENGINEER

Nahla Nabil

Bahrain · idea, satellite pipeline, tide-aware model, validation, dashboard, business case. AI/LLM engineer; NASA Space Apps local lead.

WHAT WE ARE ASKING FOR

Mentor feedback on hyperspectral reading of Gulf mangroves and sabkha vegetation, and a path into incubation to validate with a **real restoration partner**.

nahla-nabil.github.io/blue-carbon-guardian · github.com/Nahla-Nabil/blue-carbon-guardian